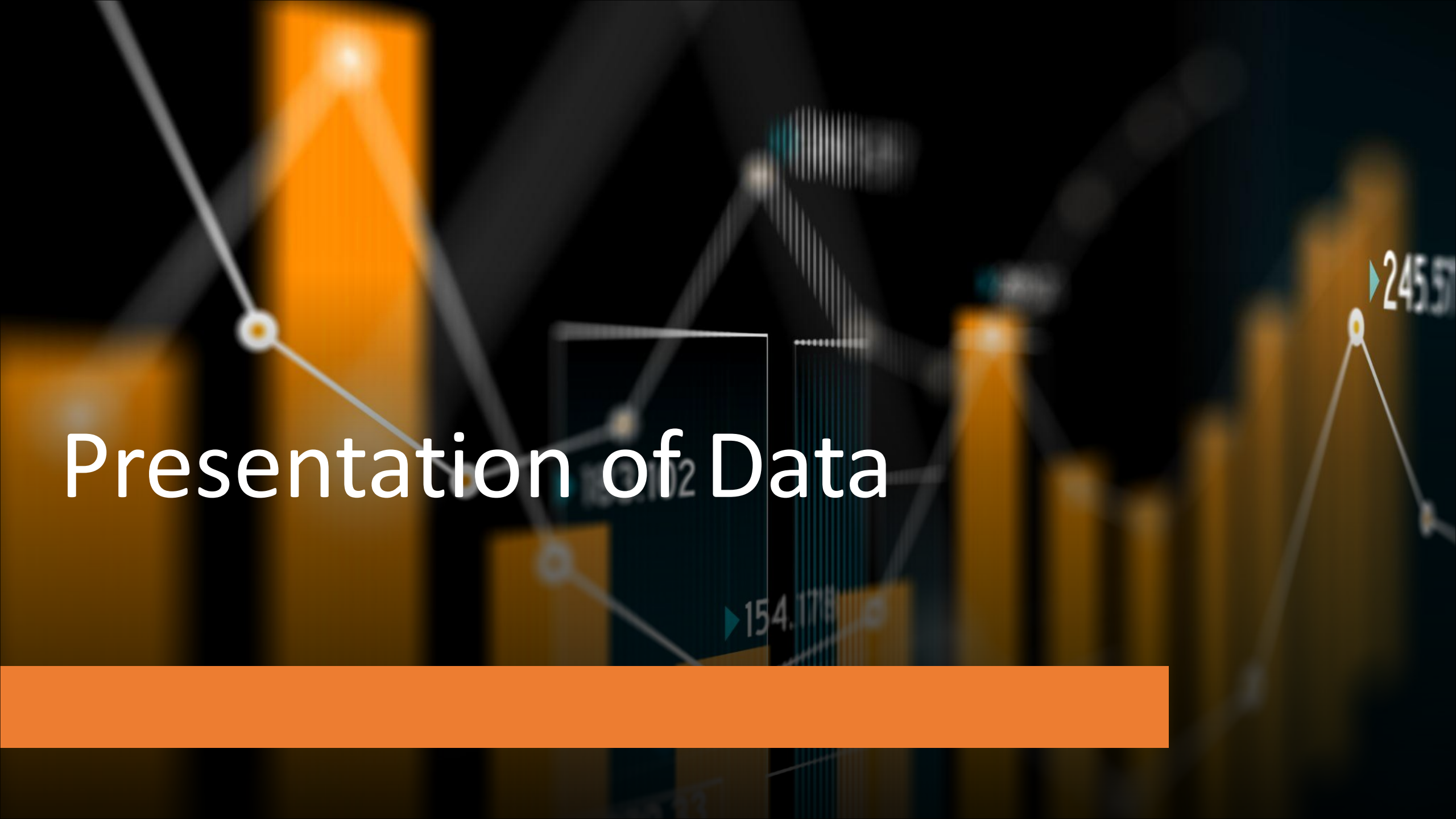


The background features a dark blue field with various abstract elements. There are several orange and yellow rectangular blocks of varying sizes, some of which are blurred. A network of thin white and light blue lines connects small circular nodes, creating a web-like structure. Some of these lines are thicker and more prominent. Faint numerical values like '245.5', '154.178', and '163.102' are visible in the background, suggesting a data visualization theme.

Presentation of Data

Frequency Distribution

A frequency distribution table is a table that displays the frequency the number of instances in which a specific value of a variable is occurring in a sample.

Univariate Frequency Distribution

When the sample data is collected corresponding to single variable which for further statistical analysis is represented in frequency distribution table is referred to as univariate frequency distribution.

- **Frequency Distribution for Discrete Variable**
- **Frequency Distribution for Continuous Variable**

Frequency Distribution for Discrete Variable

In this case the values collected on a characteristic are discrete i.e. countable. To identify the span of variation in the data we find maximum and minimum values from the data

- a) If the span is less, we list the values in one column
- b) If the span is more we group data into mutually exclusive and exhaustive classes in the first column

Frequency Distribution for Continuous Variable

Continuous set of class intervals 0-10,10-20,20-3 etc. We use such classes for continuous variates. These class intervals are called exclusive class intervals because the upper end point of each class is not included in that class while calculating class frequency. The two end points of each class interval are termed as lower and upper class boundary.

Frequency Distribution For Discontinuous Variable

Discontinuous set of class intervals 0-9,10-19,20-29 etc. We use such class intervals for discrete variables where range of values is large. These class intervals are called inclusive class intervals because both the end values of each class are included in that class interval while calculating the class frequency. The two end points of each class-interval are termed as lower and upper class limits

While Constructing a frequency table using discontinuous or continuous class intervals we need to decide

1. How many classes would be ideal?
2. What should be width of each of these classes?

$$\mathbf{k=1+3.322 \log_{10}n}$$

k = Number of classes

n = No of observations

$$\mathbf{Width = \frac{Maximum\ Observation - Minimum\ Observation}{Number\ of\ classes}}$$

Examples:

1. Suppose we have a set of 30 observations n number of children in 30 families as given below:

3,0,0,1,3,2,1,0,4,2,3,3,0,1,3,2,1,4,3,2,0,1,4,2,1,1,1,3,2,2,

Prepare a frequency table for this data

SOLUTION:

Since the number of children is between 0 and 4 We can use discrete set of class 0,1,2,3,4.

2. The marks obtained by 50 students in an examination are given below

87,175,156,149,163,148,128,112,124,136,168,164,155,127,120,102,88,95,110,125,140,128,136,154,172,135,128,108,168,172,149,137,126,118,105,95,85,165,170,150,80,130,90,178,160,140,125,120,100,110.

Prepare a frequency table for the marks obtained by the students

$N = 50$

$K = 1 + 3.322 \log_{10} N = 1 + 3.322 \log_{10} 50 = 6.6439 = 7$

Width = $\frac{\text{Max. obs.} - \text{Min. obs.}}{K} = \frac{178 - 80}{7} = 14$

which can be rounded to 15.

∴ The class intervals will be taken as 80-94, 95-109, 110-124,

Marks	Tally Marks	No. of Students
80-94		5
95-109		6
110-124		7
125-139		12
140-154		7
155-169		8
170-184		5
	Total	50

3. Following data is about rainfall in the month of July at a certain place Prepare a frequency table taking class intervals as (40-50),(50-60),.....

57.6,72.8,48.1,71.4,83.1,91.6,71.3,63.4,43.9,69.2,87.5,90.1,98.8,49.2,54.6,71.5,62.7,59.7,48.3,54.1,73.6,48.2,54.6,77.1,49.6,58.3,60.5,63.2,54.7,65,70.1

Rainfall (in mms)	Tally Marks	No. of days
40-50		6
50-60		7
60-70		6
70-80		7
80-90		2
90-100		3
	Total	31

Relative Frequency

Relative frequency for a class interval is the ratio of frequency of that class interval(f) to the total frequency (N) i.e. **Relative frequency** = f / N

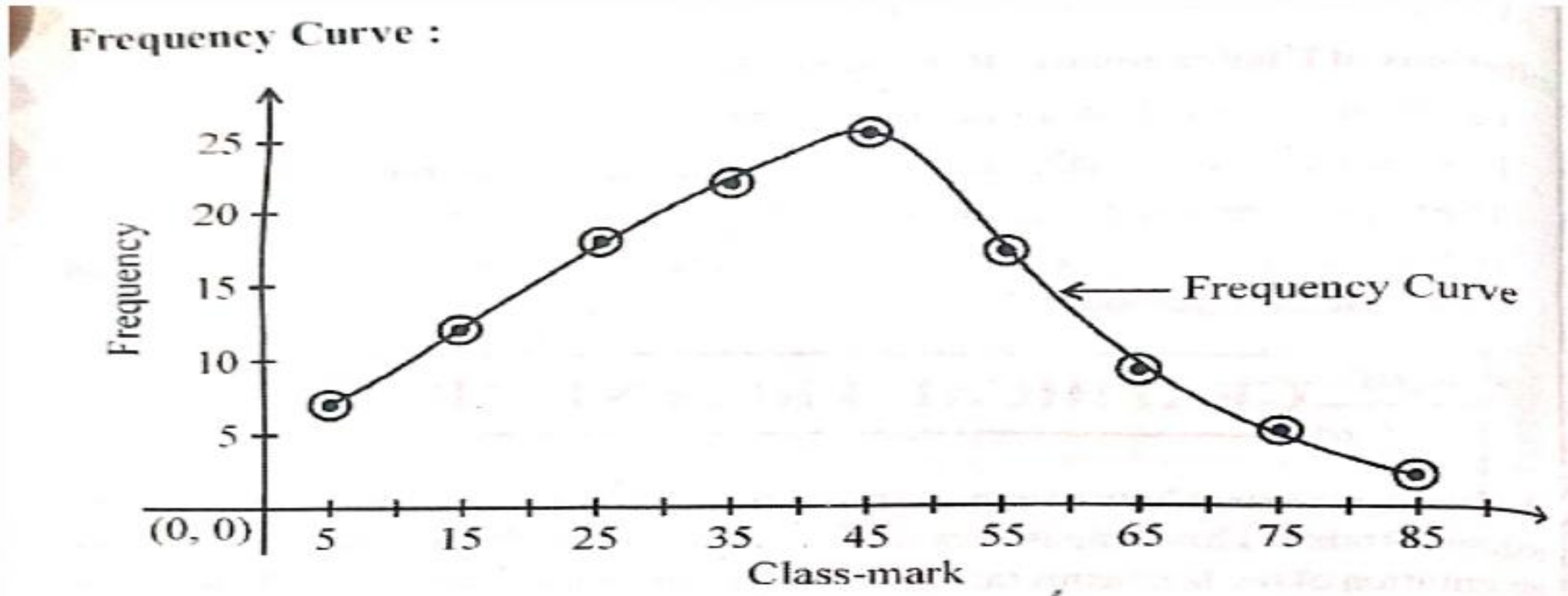
The following is record of profits of a shop in 50 days. Calculate Relative frequencies. State the class boundary for the fourth class interval. Find the number of days when profit is less than 9000

FREQUENCY CURVE AND POLYGON

- To construct frequency curve we take value of observed variable x on the X-axis and the frequency (f) along the Y-axis.
- Points are joined by a smooth curve to obtain frequency curve for discrete frequency distribution.
- If frequency distribution is in the form of class intervals then x will be the class mark of the class interval.
- For frequency polygon points are joined by straight lines and end points are joined to the X axis

Draw the frequency curve for the following data

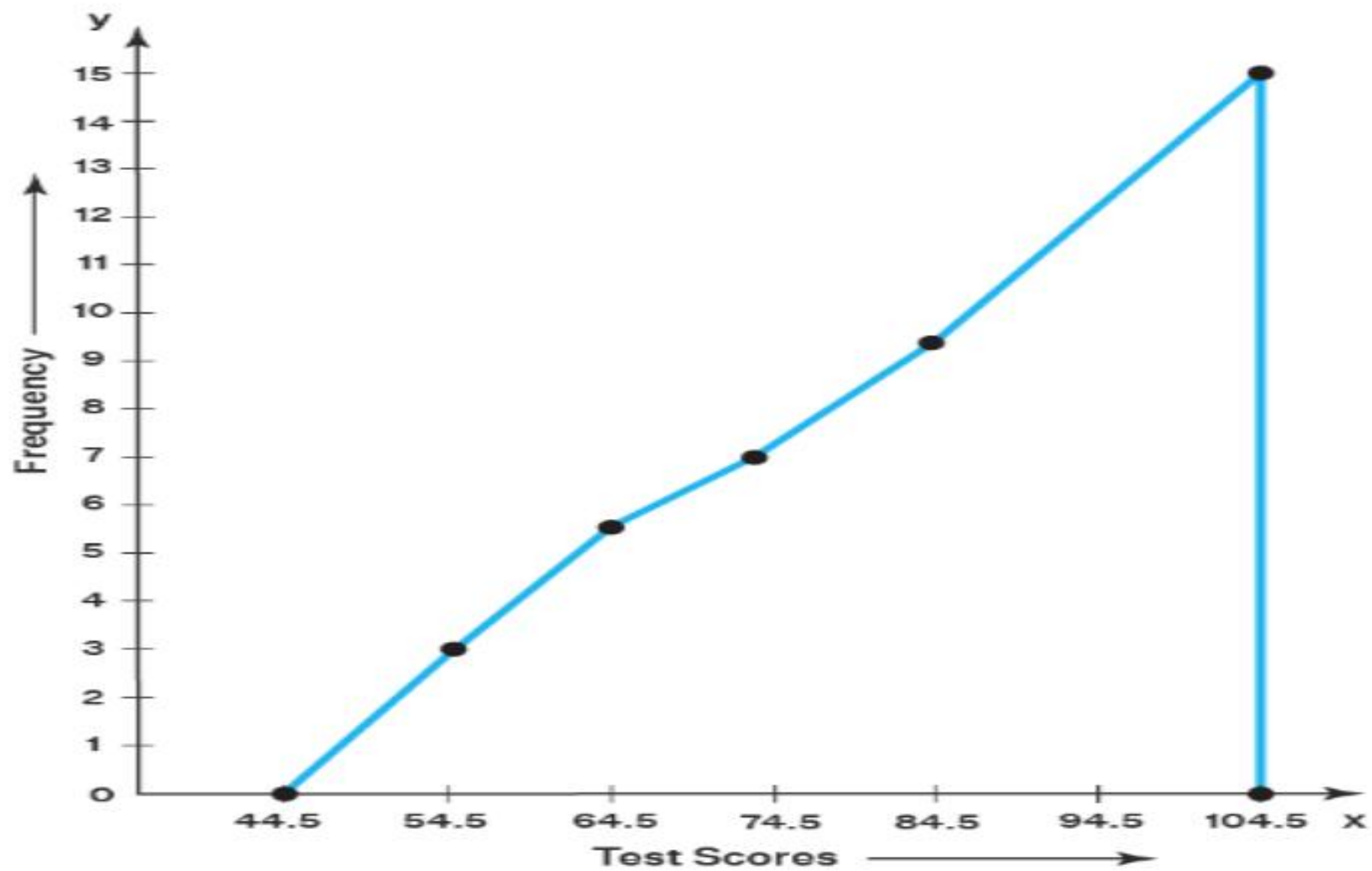
Marks	0-10	10-20	20-30	30-40	40-50	50-60	60-70	70-80	80-90
No of Students	7	12	18	22	25	17	9	5	2



Construct a frequency polygon using the data given below.

Test Scores	Frequency
49.5 - 59.5	10
59.5 - 69.5	3
69.5 - 79.5	7
79.5 - 89.5	15
89.5 - 99.5	5

Test Scores	Frequency	Classmark
39.5-49.5	0	0
49.5 - 59.5	3	54.5
59.5 - 69.5	5	64.5
69.5 - 79.5	7	74.5
79.5 - 89.5	10	84.5
89.5 - 99.5	15	94.5
99.5- 109.5	0	0



HISTOGRAM:

A histogram is a graph containing adjacent rectangles with class-interval on X-axis and corresponding frequency on Y-axis being the height of the rectangle, provided the class intervals are of same width

If class intervals are of unequal width height of the rectangle(Y axis) is taken as frequency density of each class interval

$$\text{Frequency density} = \frac{\text{Frequency}}{\text{Width}}$$

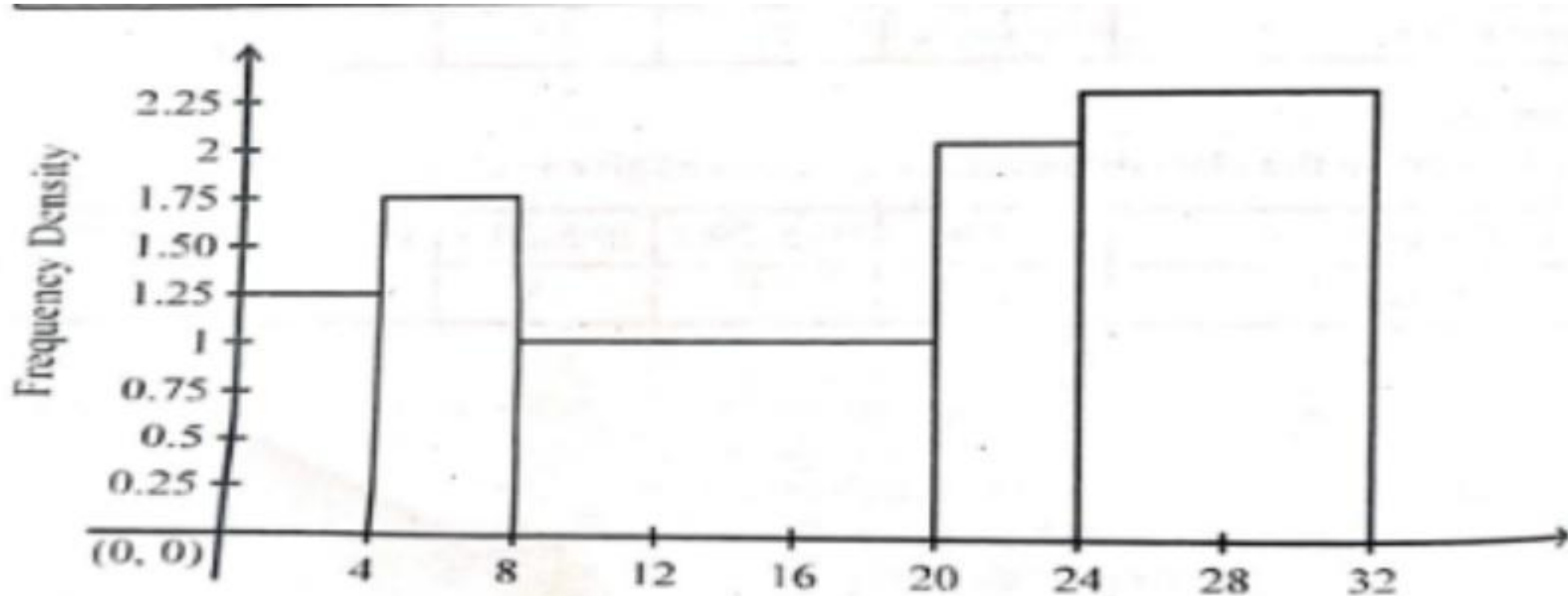
Draw histogram for the following data

Q.1

Class Interval	10-19	20-29	30-39	40-49	50-59
Frequency	6	11	15	10	5

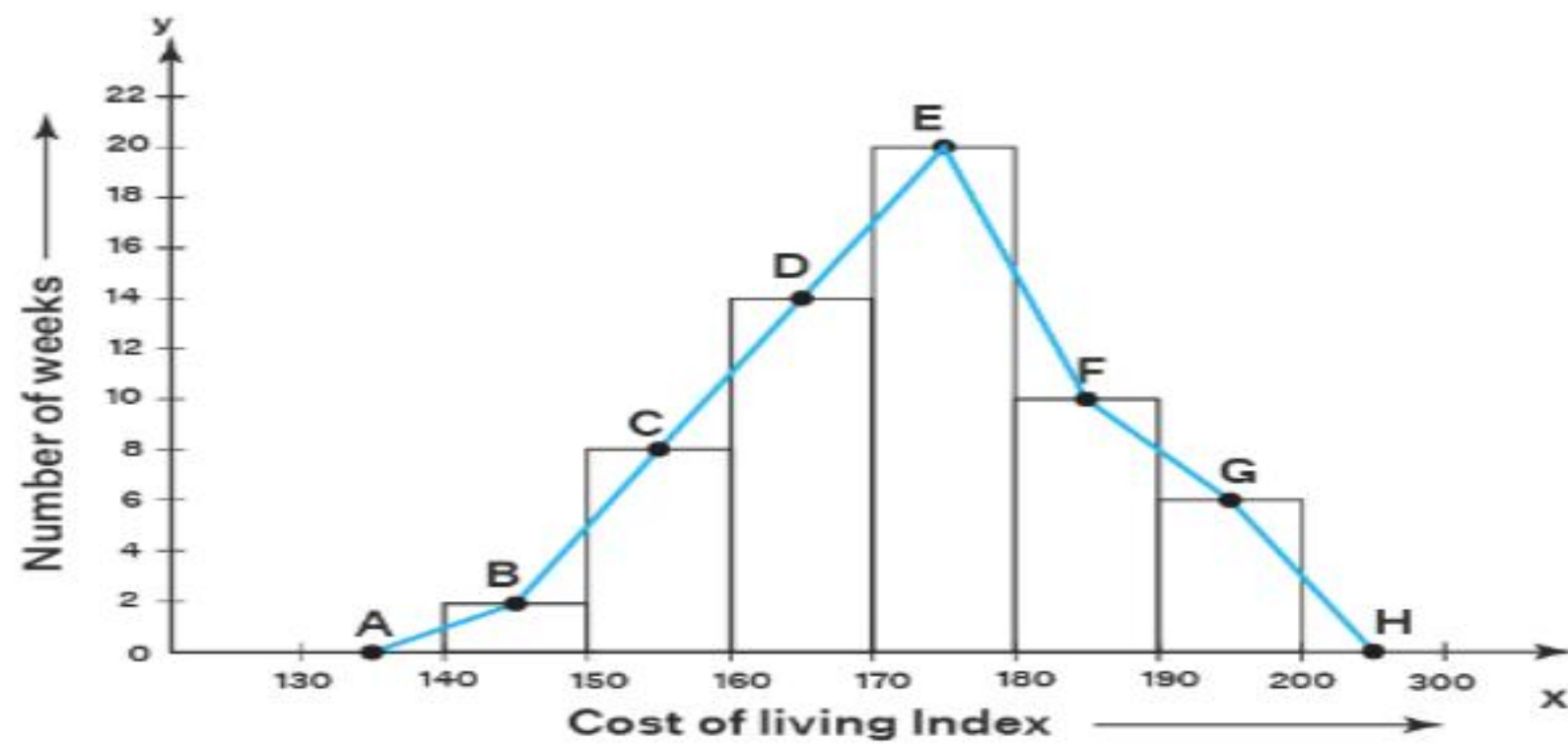
Q.2

Class Interval	0-4	4-8	8-20	20-24	24-32
Frequency	5	7	12	8	18



In a city, the weekly observations made in a study on the cost of a living index are given in the following table: Draw a frequency polygon for the data below with a histogram.

Cost of Living Index	Number of weeks
140 - 150	2
150 - 160	8
160 - 170	14
170 - 180	20
180 - 190	10
190 - 200	6
Total	60



OGIVES OR CUMULATIVE FREQUENCY CURVES

Ogives are one more way of presenting continuous frequency distribution.

Ogives or Cumulative Frequencies are of two types

- i. Less than type ogive
- ii. More than or equal to type ogive

Less Than Type Ogive

- The upper class boundaries are taken on X-axis and less than type cumulative frequency on Y axis
- Less than type cumulative frequency of the first class interval is frequency of that class interval.
- Less than type cumulative frequency of the remaining class intervals are calculated by adding cumulative frequency of the previous class interval
- Less than type cumulative frequency represents the number of observations less than the upper class boundary of the class interval

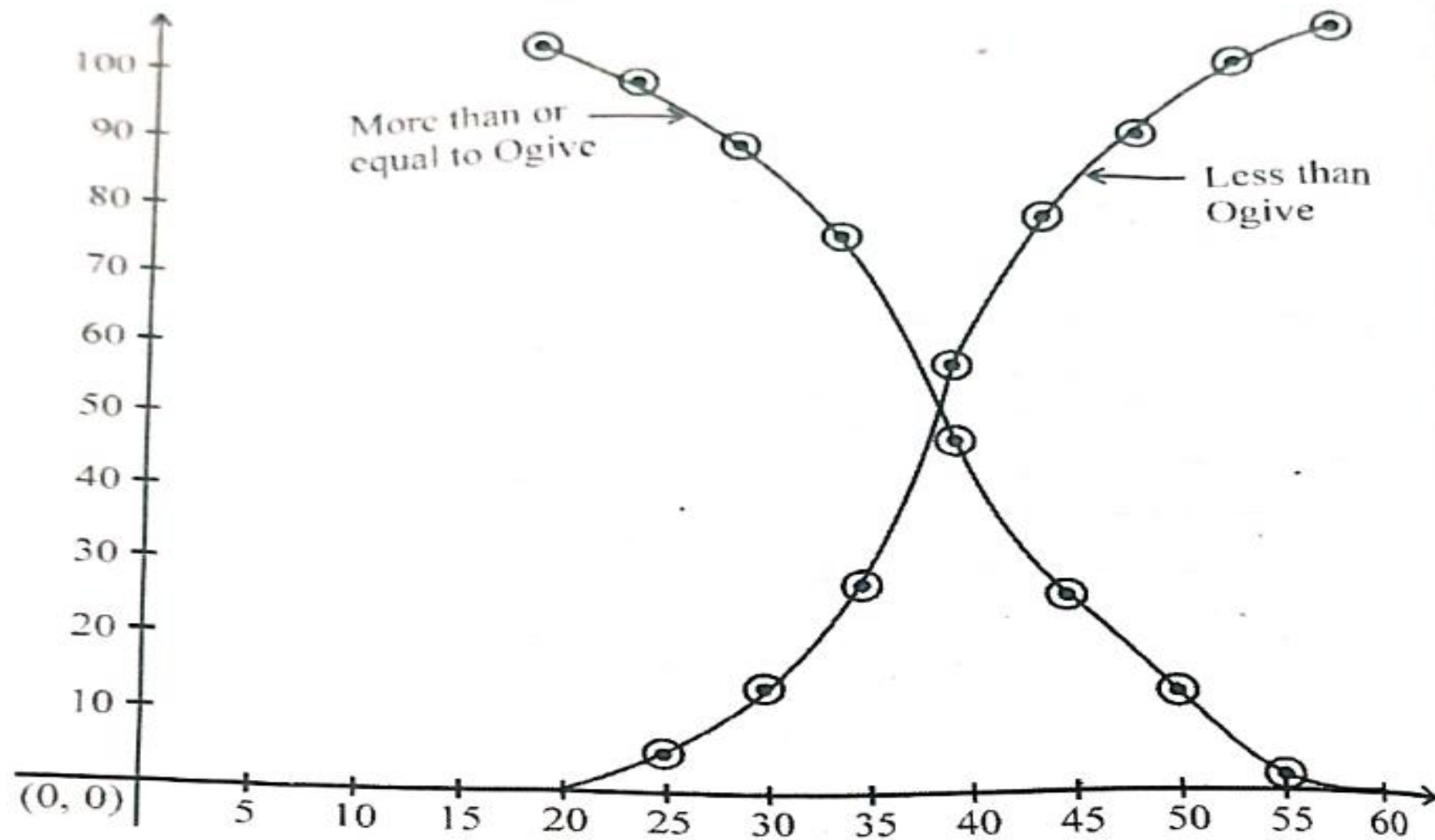
More than or equal to type ogive

- The lower class boundaries are taken along X-axis and more than or equal to cumulative frequency on Y-axis.
- More than or equal to cumulative frequency of the first class is total frequency.
- For the remaining class they are calculated by subtracting frequency of the earlier class interval from the cumulative frequency of the earlier class interval
- More than or equal to cumulative frequencies represent the number of observations more than or equal to lower class boundary of the class interval

Examples

1. The following is the distribution of age (in years) of 100 employees of a company:

Age(in years)	20-25	25-30	30-35	35-40	40-45	45-50	50-55	55-60
No of Employees	5	9	13	28	20	12	10	3

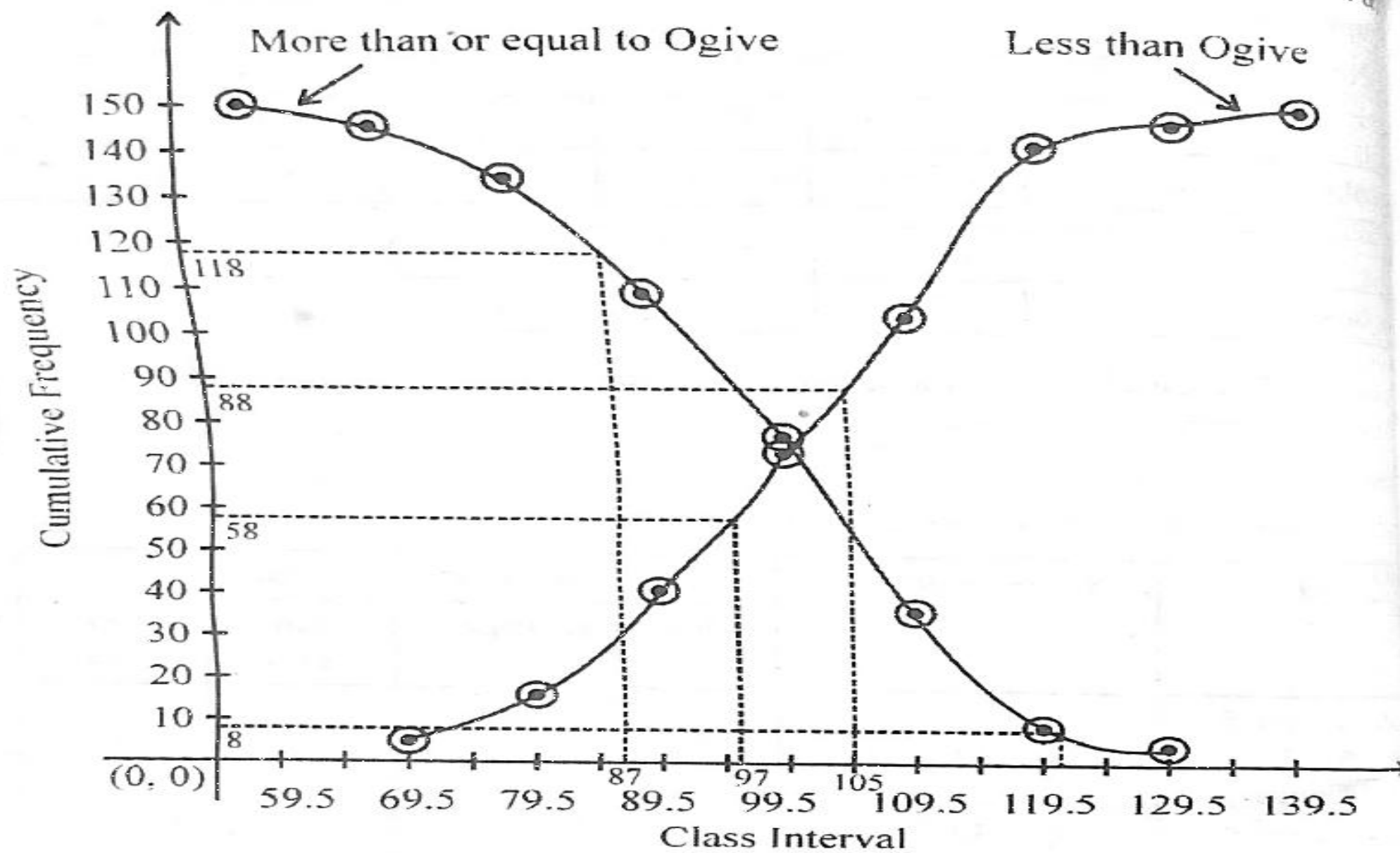


2. Draw the Ogives for the following IQ's of 150 students
Hence state the number of student with IQ

- i. Less than 105
- ii. More than or equal to 87
- iii. Between 97 and 122

IQ	60-69	70-79	80-89	90-99	100-109	110-119	120-129	130-139
No of students	5	11	25	32	41	27	5	4

IQ	No. of Students	Cumulative frequency	
		less than type	more than or equal to type
59.5 - 69.5	5	5	150
69.5 - 79.5	11	16	145
79.5 - 89.5	25	41	134
89.5 - 99.5	32	73	109
99.5 - 109.5	41	114	77
109.5 - 119.5	27	141	36
119.5 - 129.5	5	146	9
129.5 - 139.5	4	150	4



Stem and Leaf Plot

- Stem and leaf plot represent ungrouped data
 - Ungrouped data are represented by a frequency table but once we have frequency table we loss some information on how the observations are distributed in a class interval
 - The loss of information is avoided in stem and leaf displays
 - It gives rank order of the item in the data set and shape of distribution
-
- **Steps for drawing stem and leaf plot**
 1. Divide each observation into two parts. One or more consists of one or more leading digits as stem and the rest of the digits as leaf.
eg : For a 3 digit number , the first 2 digits form the stem and last digit form the leaf
 2. The stem values are listed out to the left of a vertical line and each leaf value corresponding to a stem is written in a horizontal line to the right of the stem
 3. Finally , we arrange all the leaves in each row in ascending order

Examples :

1. Present the following data in a stem and leaf display : 47,53,68,57,66,48,52,43

2. Arrange the data below using a stem and leaf display :

57,94,84,63,72,51,50,48,61,71,93,82,100,89,67,78,79,78,76,85,87,73,66,99,84,72,66

Check Sheet

- A check sheet is a data collection tool designed to help you gather and organize data efficiently.
- Its primary purpose is to provide a structured way of recording information about the occurrence or frequency of events, defects, or items of interest.
- It is also known as frequency table.
- Check sheets are most commonly used for categorical or discrete data, used for counting the number of occurrences or responses falling into specific categories.
- Examples : Defects in a manufacturing process, types of customer complaints, or website user actions (e.g., clicks on different links).

Types of Check Sheets:

There are various types of check sheets designed for specific purposes, including:

Frequency Check Sheet: Used to count the number of occurrences of specific events or defects over time.

Location Check Sheet: Used to record the locations where certain events or issues occur, helping identify patterns in spatial data.

Time Check Sheet: Used to track events or activities over time, helping identify trends or time-based patterns.

Cause-and-Effect Check Sheet: Used to record potential causes or contributing factors related to a specific problem or outcome.

Example: Defect Tracking in Manufacturing

Suppose you work in a manufacturing facility that produces electronic devices, and you want to track the types of defects that occur during the production process.

Defect Types	Monday	Tuesday	Wednesday	Thursday	Friday
Component	3	2	1	4	2
Wiring	1	0	1	0	2
Assembly	2	1	0	1	1
Packaging	0	1	0	2	0

Example: Accident Location Tracking in Construction

Suppose you are a safety manager at a construction site, and you want to track the locations where accidents and safety incidents are occurring.

Accident Locations	Location A	Location B	Location C	Location D
Monday	1	2	1	1
Tuesday	0	0	0	2
Wednesday	0	2	0	1
Thursday	2	1	2	1
Friday	0	1	1	1

Example: Patient Vital Sign Monitoring

Suppose you work in a hospital as a nurse, and you are responsible for monitoring the vital signs of patients in the Intensive Care Unit (ICU). You want to ensure that you regularly check and record patients' vital signs at specified time intervals.

	8:00 am	10:00 am	12:00 pm	2:00 pm	4:00 pm
Patient 1	X	X	X	X	X
Patient 2	X	X	X	X	X
Patient 3	X	X	X	X	X
Patient 4	X	X	X	X	X

Suppose you are the manager of a restaurant, and you want to understand the various factors or causes that contribute to customer complaints.

****Major Categories of Causes:****

- ☐ Food Quality
- ☐ Service
- ☐ Cleanliness
- ☐ Wait Time
- ☐ Ambience
- ☐ Other (Specify): _____

****Detailed Causes within Major Categories:****

[Food Quality]

- ☐ Overcooked
- ☐ Undercooked
- ☐ Wrong Order
- ☐ Taste
- ☐ Presentation
- ☐ Other (Specify): _____

[Service]

- ☐ Slow Service
- ☐ Rude Staff
- ☐ Mistakes in Orders
- ☐ Lack of Attention
- ☐ Waiting Time
- ☐ Other (Specify): _____

[Cleanliness]

- ☐ Dirty Tables
- ☐ Dirty Restrooms
- ☐ Unclean Utensils
- ☐ Unkempt Floors
- ☐ Other (Specify): _____

[Wait Time]

- ☐ Long Wait Times
- ☐ Delayed Orders
- ☐ Slow Checkout
- ☐ Other (Specify): _____

Pareto Chart

1. Pareto chart is a type of data visualization tool used to identify and prioritize the most significant factors contributing to a problem or a set of issues.
2. It's a combination of a bar chart and a line graph, and it follows the Pareto principle, also known as the 80/20 rule, which states that roughly 80% of problem is just caused by 20% of few major factors referred to as vital few remaining 20% problem is caused by 80% of minor factors referred to as trivial many
3. The Pareto chart helps you focus your attention on the "vital few" factors that contribute the most to a problem, enabling you to prioritize efforts and allocate resources efficiently. It's a powerful tool for quality improvement, problem-solving, and decision-making.

In a Pareto chart:

- The x-axis represents categories or items that are being analyzed.
- The y-axis represents the frequency, count, or impact of each category.

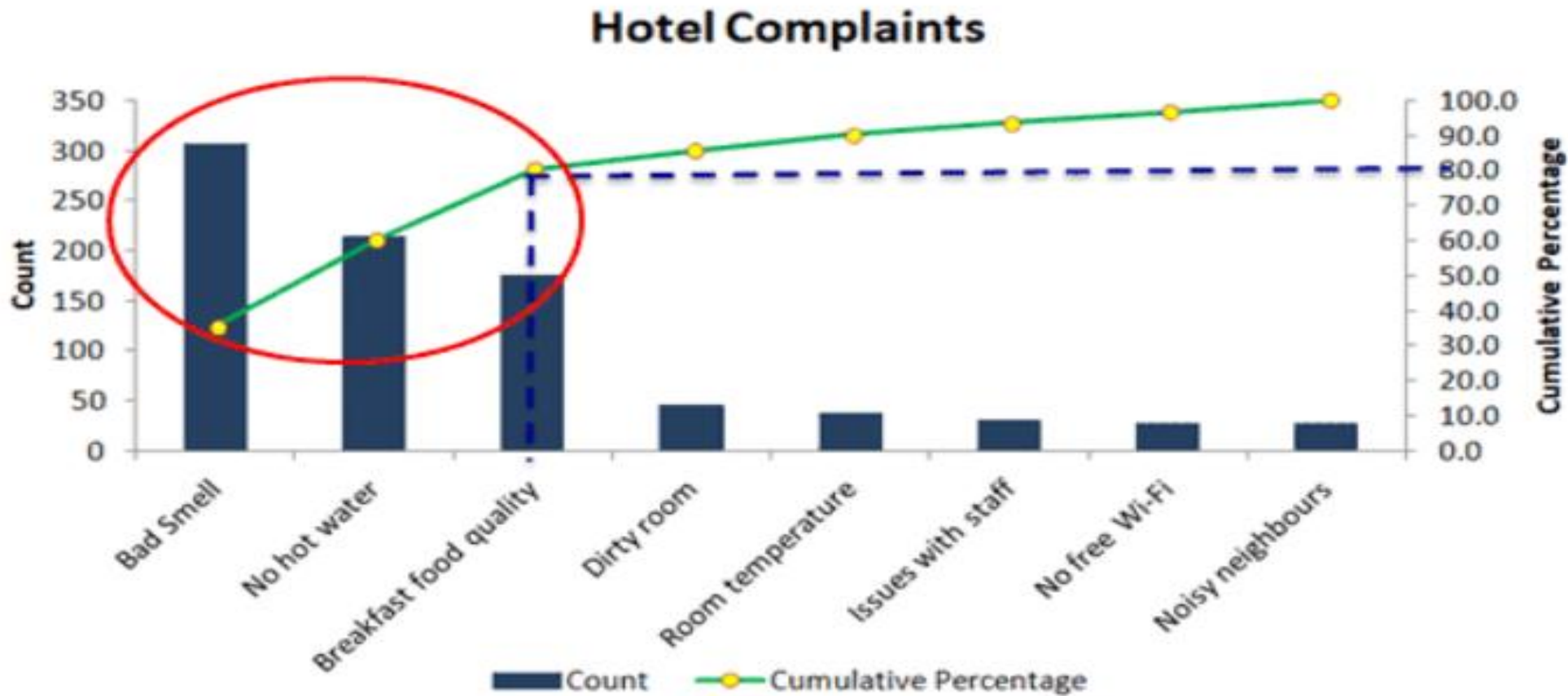
Steps to draw Pareto Chart

1. The X-axis represents the different categories and Y-axis represent the frequencies(count)
2. For each category, draw a vertical bar on the chart. The height of each bar corresponds to the frequency of that category. Make sure to label each category on the x-axis.
3. Plot the cumulative percentages on the same chart using a line graph.
4. Start at the bottom left corner (0% on the y-axis) and connect the points to form a line that rises towards the top right corner (100% on the y-axis).

Example : Customer Complaints in a Restaurant

Suppose you manage a restaurant, and you want to identify the main reasons for customer complaints.

Complaint type	Count	Cumula	Cumulative Percentage
Bad Smell	307	307	35.409
No hot water	215	522	60.208
Breakfast food	175	697	80.392
Dirty Room	46	743	85.698
Room Temp	38	781	90.081
Issues with Sta	30	811	93.541
No free Wi-fi	28	839	96.770
Noisy Neighbou	28	867	100.000



The above diagram shows that the top three categories (bad smell, no hot water, and breakfast food quality) contribute to 80% of customer complaints.